Emergency Triage, Red Flags, Cervical Screen & Subjective

A · PATIENT DEMOGRAPHICS & SHOULDER COMPLEX PRESENTATION PROFILE

INTAKE

Patient Name _____ Age _____ Sex ☐ M ☐ F Date _____

Occupation / Overhead Sport _____ Dominant Arm ☐ R ☐ L Affected Side ☐ R ☐ L ☐ Bilat Onset ☐ Traumatic ☐ Insidious

NPRS Now (circle) 0 1 2 3 4 5 6 7 8 9 10 Night Pain ☐ No ☐ Yes Best _____ Worst _____

B · BESS/BOA RED-FLAG TRIAGE, DISLOCATION & SYSTEMIC SCREENING

TRIAGE GATE

BESS/BOA PATHWAY RED FLAGS (KULKARNI 2015)

BESS/BOA

- ☐ **Infection:** **"A suspected infected joint needs same day emergency referral."** (Guideline)
- ☐ **Dislocation:** **"An unreduced dislocation needs same day emergency referral."** (Guideline)
- ☐ **Tumour / malignancy:** **"Suspected tumour and malignancy will need urgent referral following the local 2-week cancer referral pathway."** (Guideline)
- ☐ **Traumatic cuff tear:** **"An acute cuff tear as a result of a traumatic event needs urgent referral and ideally should be seen in the next available outpatient clinic."** (Guideline)

SEPTIC ARTHRITIS & POSTERIOR DISLOCATION

Emergency

- ☐ **Septic arthritis — synovial fluid** (Margaretten 2007 JAMA) [M]: **WBC > 25,000/µL: LR 2.9 (2.5–3.4); > 50,000/µL: LR 7.7 (5.7–11.0)**
- ☐ **Posterior dislocation** (Xu 2015 J Orthop Translat) [L]: **rare (< 4% of all shoulder dislocations); initial diagnosis missed in 150 patients (73.2%); AP radiograph alone gave the right diagnosis in only 19 of 166 (11.4%); axillary / Y view / CT confirmed all 205**

TRAUMATIC ANTERIOR DISLOCATION ASSOCIATED INJURIES

Brownson 2015

- ☐ **Associated injury rates** (Brownson 2015 BESS/BOA) (Guideline): **clinically apparent neurological injury ≈ 13.5% of dislocations (axillary nerve two-thirds); EMG shows nerve injury in 48%; greater-tuberosity fracture ≈ 16%; rotator cuff tears 41% (40–55 y), 71% (56–70 y), 100% (> 70 y)**
- ☐ **Nerve injury follow-up** (Brownson 2015) (Guideline): **"EMG studies should be performed at 6 weeks if no clinical recovery is seen"** → nerve-injury centre referral if no recovery

PANCOAST, PMR, VISCERAL & CERVICAL SCREEN

Systemic & Neck

- ☐ **Pancoast tumour** (Bisbinas 1999 Ann R Coll Surg Engl) [L]: **4 cases presenting to orthopaedics: mean diagnostic delay 18.5 months**
- ☐ **Polymyalgia rheumatica (EULAR/ACR 2012)** (Dasgupta 2012 Ann Rheum Dis) [M]: **score ≥ 4 (no ultrasound): Sn 68%, Sp 78%; score ≥ 5 with ultrasound: Sn 66%, Sp 81%**
- ☐ **Cardiac / visceral referral:** myocardial ischaemia, diaphragmatic, hepatobiliary or pulmonary referral (non-mechanical shoulder pain) [E]
- ☐ **Cervical radiculopathy screen** (Wainner 2003 Spine) [M]: **4-item test cluster LR+ 30.3 (wide CI); ULTT A ▸ most useful to rule out**

TRIAGE DISPOSITION: ☐ **Cleared — proceed with clinical dossier** ☐ Next available outpatient clinic (acute cuff tear) ☐ 2-week cancer referral pathway ☐ **Same day emergency referral (infected joint / unreduced dislocation)**

C · CLINICAL PATHWAY ROUTER & RAPID CERVICAL / SCAPULAR SCREEN

RAPID

CLINICAL PATHWAY ROUTER

Tick route → jump to matching pages

☐ A · CUFF: 1 → 3 → 4 → 8 → 9 ☐ B · FROZEN / RCRSP: 1 → 2 → 3 → 6 → 7 → 8 → 9 ☐ C · POST-OP: 1 → 3 → 5 → 8 → 9

D · SUBJECTIVE HISTORY, SYMPTOM BEHAVIOUR & SHOULDER / NECK / ARM PAIN MAP

SUBJECTIVE

SYMPTOM LOCATION & CLINICAL PATTERN

Clinical Signpost

- ☐ **Anterolateral / Deltoid Pain on Overhead Activity:** full passive ER → **Subacromial pain / RCRSP (p2/p4)**
- ☐ **Global Stiffness & Severe Night Pain:** passive ER loss → **Frozen shoulder / GH OA (p2/p6)**
- ☐ **Superior AC Joint Pain & High-Arc Provocation:** local tenderness → **AC joint pathology (p2/p4)**
- ☐ **Apprehension or Slipping in Abduction + ER:** trauma or hyperlaxity → **Traumatic or Stanmore II/III instability (p2/p4)**
- ☐ **Neck Pain Radiating Distal to Elbow with Paresthesia:** (+) Wainner cluster / ULTT A ▸ → **Cervical radiculopathy (p1/p3)**

AGGRAVATING, EASING & IRRITABILITY PROFILE

Behaviour

Aggravating _____ Easing _____

Irritability: ☐ Low ☐ Moderate ☐ High (rest/night pain)

MECHANISM & 24-HOUR PATTERN

History

Mechanism: ☐ Fall / FOOSH ☐ Dislocation / subluxation ☐ Overuse / overhead load ☐ Insidious

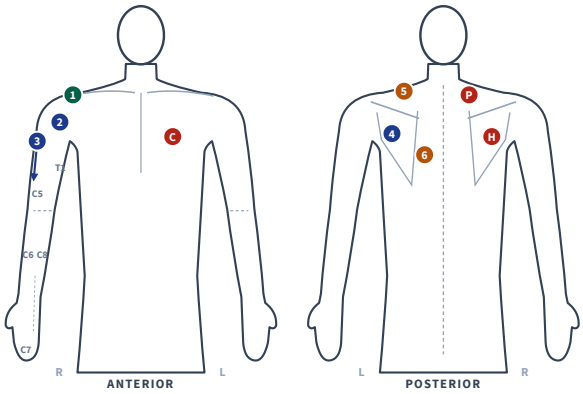
24 h: ☐ Morning stiffness ☐ Worse with activity ☐ Night pain lying on side ☐ Constant / unremitting

Previous episodes / treatment: _____

Hand dominance & sport / work demand: _____

SHOULDER, NECK & ARM BODY CHART

Clinical signposts [E]



Mark: shade = pain · //// = pins & needles · ×× = numbness · → = radiation · write 0–10 beside each area

1. **AC joint** — points to top of shoulder
2. **Anterior shoulder** — biceps groove / anterior instability
3. **Lateral deltoid → insertion** — cuff / subacromial referral
4. **Posterior shoulder** — posterior cuff / suprascapular n.
5. **Upper trapezius / suprascapular** — neck referral
6. **Medial scapular border** — neck referral

C. **Left chest / arm / jaw** — cardiac

P. **Apex, medial scapula + ulnar arm** — Pancoast

H. **Right shoulder tip / scapula** — hepatobiliary / diaphragm

Pain below the elbow, paraesthesia or pain changed by neck movement → cervical screen (Wainner). **Pain not changed by shoulder movement** → red-flag screen (B).

SUBJECTIVE HISTORY · MECHANISM · PATIENT GOALS (PSFS)

Clinician _____ Date _____

Differential & stiff-vs-mobile stratification: first compare passive external rotation (Cadogan 2016 / Kelley 2013), classify instability (Brownson 2015 / Noorani 2019), and clear the cervical spine and thoracic outlet.

1 · RCRSP, ROTATOR CUFF TEAR, FROZEN SHOULDER & GH OA — MATRIX A

- ☐ **Subacromial pain — definition** Kulkarni 2015 (Guideline): **"commonly located to the top and lateral side of the shoulder … exaggerated by overhead activity … can be associated with night pain … usually associated with full passive range of movement"**
- ☐ **RCRSP terminology & burden:** umbrella term for subacromial pain, RC tendinopathy, partial and full-thickness tears; exercise ≈ surgery (Lewis 2016 Man Ther) [E] **• 2.4% of UK primary care consultations; subacromial pain up to 70% of shoulder-pain problems** (Beard 2018 Lancet CSAW) [M]
- ☐ **Full-thickness rotator cuff tear: "overall prevalence of rotator cuff tears is 34% and … risk increases significantly with age"** (Kulkarni 2015, Guideline); US cluster **Sp 98% (94–100), LR+ 6.2 (1.5–25.7)** (Cadogan 2016) [M]
- ☐ **Frozen shoulder (adhesive capsulitis):** global active & passive restriction; **ROM loss > 25% in at least 2 planes AND passive ER loss > 50% of the other side or < 30° ER** (Kelley 2013 JOSPT CPG, Guideline)
- ☐ **Glenohumeral osteoarthritis (GH OA): those with ER loss had more GH arthropathy on X-ray (17% vs 3%)** (Cadogan 2016 PLoS One) [M]

2 · INSTABILITY, AC JOINT, SLAP/BICEPS, CALCIFIC & CERVICAL/TOS — MATRIX B

- ☐ **Traumatic instability — recurrence** Brownson 2015 (Guideline): **almost 90% of recurrences within 2 years; males under 20 ≈ 72% chance of recurrent instability; immobilisation > 1 week does not reduce recurrence**
- ☐ **Atraumatic instability (Stanmore II/III)** Noorani 2019 (Guideline): **Stanmore polar types II (hyperlaxity) and III (muscle patterning, structurally normal)**
- ☐ **AC joint pathology:** superior AC pain & high-arc provocation; **no impingement or AC-joint test showed significant diagnostic accuracy (2008 meta-analysis)** (Hegedus 2008 BJSM) [M]
- ☐ **SLAP / long head of biceps:** deep mechanical clicking/pain; **pooled DOR of physical tests 1.38 (1.13–1.69) — poor** (Gismervik 2017 BMC Musculoskelet Disord) [M]
- ☐ **Calcific tendinopathy:** acute or chronic calcific tendinitis of the rotator cuff with severe flare pain (see ESWT & barbotage on p6) [E]
- ☐ **Cervical radiculopathy / Thoracic Outlet Syndrome (TOS):** cervical screen (Wainner cluster, p1) [M] and TOS neurovascular upper-limb symptoms [E]

3 · FROZEN SHOULDER BOX (KELLEY 2013 JOSPT CPG) — GUIDELINE

Kelley 2013

- ☐ **Definition used** (Guideline): **ROM loss > 25% in at least 2 planes AND passive ER loss > 50% of the other side or < 30° ER**
- ☐ **Prevalence & risk factors** (Guideline): **shoulder pain 2.4–26%; primary adhesive capsulitis 2% to 5.3%; secondary (diabetes/thyroid) 4.3%–38%; diabetes in women 23.7% vs 4.7% and men 38.0% vs 6.5%; hypothyroidism (women) 21.1% vs 7.9%; age 40–65, peak 51–55; contralateral risk 5%–34%, bilateral simultaneous up to 14%**
- ☐ **4 clinical stages** (Guideline): **stage 1 up to 3 months; stage 2 "freezing" 3–9 months; stage 3 "frozen" 9–15 months; stage 4 "thawing" 15–24 months; mild symptoms may persist for years**

Stage: ☐ 1 (up to 3 m) ☐ 2 (3–9 m) ☐ 3 (9–15 m) ☐ 4 (15–24 m) ☐ Diabetes / Thyroid

4 · CADOGAN PASSIVE ER RULE & CLUSTER · HANCHARD COCHRANE — PRIMARY CARE

- ☐ **Subacromial pain vs stiff shoulder** (Cadogan 2016 PLoS One) [M]: **full passive ER: Sn 99%, LR− 0.14 → loss of passive ER argues against subacromial pain; those with ER loss had more GH arthropathy on X-ray (17% vs 3%)**
- ☐ **Cadogan subacromial pain cluster** (Cadogan 2016) [M]: **anterior shoulder pain + strain injury + no symptoms at end-range ER in abduction: Sp 100% (LR+ ∞, 2.9–∞); full-thickness supraspinatus tear on US: Sp 98% (94–100), LR+ 6.2 (1.5–25.7); 34% had a positive subacromial anaesthetic response**
- ☐ **Physical tests in general (Cochrane)** (Hanchard 2013 Cochrane) [H]: **insufficient evidence to base selection of physical tests in primary care**

Passive ER Status: ☐ Full passive ER (Sn 99%) ☐ Passive ER loss (> 50% or < 30°)

5 · WORKING DIAGNOSTIC SYNTHESIS & PATHWAY CLASSIFICATION — SYNTHESIS

Primary Working Diagnosis _____ **Secondary / Scapular / Neck Driver** _____ **Stanmore Polar Type** ☐ I ☐ II ☐ III

Primary Subgrouping: ☐ RCRSP / Subacromial ☐ Full-Thickness Cuff Tear ☐ Frozen shoulder ☐ GH OA ☐ Traumatic Instability ☐ Atraumatic (Stanmore II/III) ☐ AC Joint / SLAP / Calcific

DIFFERENTIAL REASONING · PASSIVE ER GATE · IMAGING & PATHWAY NOTES

1 · SHOULDER ACTIVE & PASSIVE RANGE OF MOTION

BLANK WRITE-IN FIELDS · NO NORMS

SHOULDER MOTION	ACTIVE L	ACTIVE R	PASSIVE L	PASSIVE R	SYMPTOM REPRODUCTION & END-FEEL
Flexion / Scaption / Abduction					☐ Painful arc ☐ Capsular
External Rotation (ER) at 0°					☐ Full passive ER ☐ ER loss (> 50% or < 30°)
External Rotation (ER) at 90°					☐ Pain ☐ Apprehension
Internal Rotation (IR) at 90° / Hand-Behind-Back					☐ Symmetric ☐ GIRD / restricted
Total Rotational Motion Arc (ER + IR at 90°)					☐ Symmetric ☐ Deficit > 5° (p8)

ROM — goniometer vs digital inclinometer (Kolber 2012 IJSPT) [M]: intrarater ICC ≥ 0.94 (goniometer), ≥ 0.95 (inclinometer); limits of agreement 2°–20° → do not swap instruments between visits.

2 · SCAPULAR DYSKINESIS & SYMPTOM MODIFICATION BATTERY

MCCLURE · SAT · SRT · SSMP

☐ Scapular dyskinesis test (McClure) ▸ (McClure 2009 J Athl Train) [M]: 5 reps weighted flexion + abduction (1.4 kg or 2.3 kg); agreement 75–82%, κw 0.48–0.61 • SDT in rugby players ▸ (Singh 2021 S Afr J Sports Med) [L]: slight reliability (κ 0.16; 41%)

☐ Scapular Assistance Test (SAT) ▸ [E] & Scapular Retraction Test (SRT) ▸ [E]: manual upward rotation / posterior tilt (SAT) or scapular stabilisation (SRT) during provocative elevation or Jobe testing — no verified numbers

☐ Shoulder Symptom Modification Procedure (SSMP) [E]: systematic thoracic, scapular, and humeral head symptom modification manoeuvres during provocative movement — no verified numbers

SDT Rating: ☐ Normal ☐ Subtle ☐ Obvious

Symptom Modification: ☐ SAT (+) ☐ SRT (+) ☐ SSMP Responder

3 · ISOMETRIC STRENGTH / HHD WRITE-IN TABLE & UPPER-QUARTER PERFORMANCE

NO NORMS · CKCUEST · YBT-UQ

MUSCLE GROUP / TEST (HHD [E])	LEFT	RIGHT	SYMMETRY / RATIO	PAIN REPRODUCTION & NOTES (NO NORMS)
External Rotation (ER at 0° / 90°) [E]				
Internal Rotation (IR at 0° / 90°) [E]				
Scaption / Abduction & ER:IR Ratio [E]				

☐ Closed Kinetic Chain Upper Extremity Stability Test (CKCUEST) (Tucci 2014 BMC Musculoskelet Disord) [M]: MDC 2.05–3.91 touches; at least 3 touches = real improvement • Baseline: _____ Follow-up: _____

☐ Upper Quarter Y-Balance Test (YBT-UQ) [E]: no verified numbers — write-in reach L: _____ R: _____

4 · UPPER-LIMB NEUROLOGICAL SCREEN (C5–T1)

DERMATOME · MYOTOME · REFLEX

ROOT	DERMATOME (LIGHT TOUCH / PINPRICK)	MYOTOME (KEY MUSCLE ACTION)	REFLEX	LEFT	RIGHT
C5	Lateral arm (axillary nerve badge area) [E]	Shoulder abduction (Deltoid)	Biceps (C5–C6)		
C6	Lateral forearm, thumb & index finger [E]	Elbow flexion & wrist extension	Brachioradialis (C6)		
C7	Middle finger & central palm [E]	Elbow extension & wrist flexion	Triceps (C7)		
C8	Ring & little fingers, medial forearm [E]	Finger flexion & thumb extension	—		
T1	Medial arm & epicondylar region [E]	Finger abduction / adduction (Interossei)	—		

OBJECTIVE EXAMINATION NOTES · SCAPULAR MODIFICATION · STRENGTH OBSERVATIONS

Clinician _____ Date _____

How to read: interpret single tests in clusters (Michener 2009 ▸, Park 2005, Murrell 2001). Underline = Atlas card · ▸ = verified video · Calc = interactive calculator.

1 · IMPINGEMENT / RCRSP TESTS & CLUSTERS

MICHENER · PARK · HEGEDUS

- ☐ **Neer** ▸ (Hegedus 2012 / Michener 2009) [M]: **Sn 72%, Sp 60%; LR− 0.35 (0.12–0.97)**
- ☐ **Hawkins-Kennedy** ▸ (Hegedus 2012 / Gismervik 2017) [M]: **Sn 79%, Sp 59%** • SIS MA: **DOR 2.86 (1.14–7.17); Sn 0.58, Sp 0.67**
- ☐ **Painful arc** ▸ (Hegedus 2012 / Michener 2009) [M]: **Sn 53%, Sp 76%; LR+ 2.25 (1.33–3.81), LR− 0.38 (0.16–0.90)**
- ☐ **Empty can (Jobe)** ▸ [M] & **Full can** ▸ [E]: SIS **LR+ 3.90 (1.5–10.12)** (Michener 2009); full-thickness tear **DOR 9.24 (1.99–42.84); Sn 0.74, Sp 0.77** (Gismervik 2017)
- ☐ **ER resistance (infraspinatus)** ▸ (Michener 2009) [M]: **LR+ 4.39 (1.74–11.07), LR− 0.50 (0.28–0.89)**
- ☐ **Michener cluster** ▸ (Michener 2009) [M]: **≥ 3 of 5 positive (Neer, Hawkins-Kennedy, painful arc, empty can, ER resistance) confirms; < 3 rules out; AUC 0.79; κ 0.45–0.67 (painful arc, empty can, ER resistance), 0.39–0.40 (Neer, HK)**
- ☐ **Park clusters** (Park 2005 JBJS Am) [M]: **HK + painful arc + infraspinatus test → 95% post-test probability of any impingement; painful arc + drop-arm + infraspinatus test → 91% for full-thickness tear**

2 · CUFF TEAR LAG SIGNS, MURRELL CLUSTER & SUBSCAPULARIS

TEAR · SUBSCAPULARIS

- ☐ **Drop-arm test** ▸ (Park 2005) [M] & **Hornblower sign** ▸ [E]: posterosuperior cuff integrity screen
- ☐ **ER lag sign (ERLS)** ▸: **Sp 97% for subacromial pain (injection-confirmed), highest of all tests** (Cadogan 2016) [M]; **ERLS less sensitive but more specific than Jobe; lag size correlates with tear size** (Hertel 1996) [L]
- ☐ **IR lag sign (IRLS)** ▸ (Hertel 1996) [L]: **IRLS more sensitive than lift-off; lag size correlates with tear size**
- ☐ **Murrell cuff-tear cluster** (Murrell & Walton 2001 Lancet) [L]: **supraspinatus weakness + ER weakness + impingement sign all positive (or 2 positive and age ≥ 60) → 98% chance of cuff tear**
- ☐ **Bear-hug** ▸ (Barth 2006 Arthroscopy) [L]: **Sn 60%, Sp 91.7%**
- ☐ **Belly-press / Napoleon** ▸ (Barth 2006) [L]: **belly-press Sn 40%, Sp 97.9%; Napoleon Sn 25%, Sp 97.9%**
- ☐ **Lift-off test** (Barth 2006) [L]: **Sn 17.6%, Sp 100%**

3 · GLENOHUMERAL INSTABILITY TESTS

FARBER · LO

- ☐ **Apprehension test** ▸ (Farber 2006 JBJS Am) [M]: **Sn 72%, Sp 96%, LR 20.2 (apprehension as criterion, not pain)**
- ☐ **Relocation test** ▸ (Farber 2006) [M]: **Sn 81%, Sp 92%, LR 10.4 (apprehension as criterion, not pain)** • SLAP Sn **52%** (Hegedus 2012)
- ☐ **Surprise (release) test** (Lo 2004 Am J Sports Med) [L]: **Sn 63.89%, Sp 98.91%; all three tests positive: PPV 93.6%, NPV 71.9%**
- ☐ Anterior drawer (Farber 2006) [M]: **Sn 53%, Sp 85%, LR 3.6** • **Load-and-shift** ▸ [E] • **Sulcus sign** ▸ [E] (inferior laxity)

4 · AC JOINT & SLAP / BICEPS TESTS

CHRONOPOULOS · KIM · HOLTBY

- ☐ **Cross-body adduction** ▸ & **O'Brien active compression** ▸ (Chronopoulos 2004) [L]: **cross-body Sn 77%; AC resisted extension Sn 72%; O'Brien Sn 41%, Sp 95%; all NPV > 94%, PPV < 30%**
- ☐ **AC tenderness & Paxinos** ▸ (Walton 2004 JBJS Am) [L]: **AC tenderness Sn 96%; Paxinos Sn 79% — both low specificity**
- ☐ **Biceps load II (SLAP)** (Kim 2001) [L]: **Sn 89.7%, Sp 96.9%, κ 0.815 (single-surgeon study)**
- ☐ **Speed** ▸ & **Yergason** ▸ (Holtby 2004) [L]: **Speed Sn 32%, Sp 75%, LR+ 1.28, LR− 0.91; Yergason Sn 43%, Sp 79%, LR+ 2.05, LR− 0.72** (SLAP MA Sp **95%**, Hegedus 2012 [M])
- ☐ **Compression-rotation (SLAP)** [M]: **LR+ 2.81; passive distraction highly specific** (Hegedus 2012); **DOR 6.36 (1.41–28.59); Sp 0.89, Sn 0.43** (Gismervik 2017)

5 · VALIDATED SHOULDER OUTCOME MEASURES (PROMS DASHBOARD)

PROMS · VERIFIED MCIDS ONLY

INSTRUMENT & CALCULATOR	BASE	FOLLOW	VERIFIED RELIABILITY, MDC & MCID VALUES
SPADI Calc **(Guideline)**			**MDC90 18.1; MCID 8.0 and 13.1 points (two studies)** (Kelley 2013 JOSPT CPG)
QuickDASH / DASH Calc [H]			**QuickDASH test-retest 0.90, MCID 8.0 points** (Mintken 2009) [M] • **DASH / QuickDASH ICC 0.93 / 0.91; MDC90 10.81 / 12.85; MCID 10.83 / 15.91** (Franchignoni 2014) [M] • MA: **pooled MCID 11.00 (8.59–13.41) / 11.97 (9.60–14.33); MDC90 9.04 / 9.03** (Galardini 2024) [H] • Frozen shoulder DASH: **MDC 6.6–12.2 (weighted average 10.5); MCID 10.2** (Kelley 2013)
ASES Shoulder Score Calc [M]			**ICC 0.84; MDC 9.7; MCID 6.4 points** (Michener 2002 J Shoulder Elbow Surg)
Constant Score (cuff surgery) [M]			**MCID 10.4 points** (Kukkonen 2013 J Shoulder Elbow Surg)
Oxford Shoulder Score (OSS) Calc [M]			**MCIC 6.50 (4.41–8.61) mean-change; 8.42 predictive; "at least eight points" meaningful** (Fang 2024 Arthritis Care Res)
WOSI [M]			**instability-specific QoL tool, 21 items, 4 domains — no MCID in source** (Kirkley 1998 Am J Sports Med)
NPRS Calc [M] & **PSFS** Calc [E]			**NPRS test-retest 0.74; MCID 1.1** (Mintken 2009) • **PSFS: no shoulder-specific number in this file** [E]

SPECIAL TEST CLUSTER SYNTHESIS · BASELINE PROM SCORES · TARGET MCID GOALS

POST-OP RESTRICTIONS ARE SURGEON-DIRECTED [E] — follow the operating surgeon's individualised protocol for all movement, loading, and protection parameters.

1 · POST-OPERATIVE WOUND, NEUROLOGICAL & INFECTION RE-CHECK GATE

MANDATORY EVERY VISIT

- ☐ **Injection before cuff repair** (Forsythe 2019 JBJS Am) [M]: **injection within 1 month before arthroscopic cuff repair: infection 1.3% vs 0.8% (OR 1.7, 1.0–2.9)**
- ☐ **Suspected post-op joint infection:** **"A suspected infected joint needs same day emergency referral."** (Kulkarni 2015 BESS/BOA, Guideline) • screen for erythema, purulent drainage, fever, or escalating rest pain [E]

VISIT RE-CHECK: ☐ Portals / wound clean & dry (no infection) ☐ Axillary & distal neurovascular status intact ☐ No DVT / systemic fever signs ☐ Surgeon restrictions verified

2 · ROTATOR CUFF REPAIR PROM TIMING

KLUCZYNSKI 2015

- ☐ **Cuff repair: early vs delayed PROM (MA, 28 studies, 1,729 repairs)** (Kluczynski 2015 Am J Sports Med) [M]: **level 1 studies: retear 13.7% early vs 10.5% delayed, RR 1.30 (0.74–2.30), P = .36; tears ≤ 3 cm (TO + single-row): 18.7% vs 28.2%, RR 0.66 (0.47–0.95); tears > 5 cm (double-row): 56.4% vs 20%, RR 2.82 (1.31–6.07) → tear size matters**
- ☐ **Restrictions: POST-OP RESTRICTIONS ARE SURGEON-DIRECTED** [E]

3 · BANKART VS LATARJET & RETURN TO SPORT

HOSSEIN ZADEH · MEMON

- ☐ **Bankart vs Latarjet (MA, 21 studies, 13,176 shoulders)** (Hossein Zadeh 2024 J Shoulder Elbow Surg) [M]: **arthroscopic Bankart RR 3.08 (2.03–4.68) for recurrence/revision vs Latarjet; Rowe +4.55 for Latarjet; total complications ≈ 47% lower with Bankart (RR 0.53, 0.31–0.90)**
- ☐ **Return to sport after arthroscopic Bankart (SR, 34 studies)** (Memon 2018 J Shoulder Elbow Surg) [M]: **any sport 81% (74–87%); pre-injury level 66% (57–74%); ≈ one-third do not return to pre-injury level**
- ☐ **Restrictions: POST-OP RESTRICTIONS ARE SURGEON-DIRECTED** [E]

4 · SUBACROMIAL DECOMPRESSION & MID / PASS

CSAW · FIMPACT · KANTO

- ☐ **CSAW (n = 313)** (Beard 2018 Lancet) [H]: **OSS at 6 months: decompression 32.7 vs arthroscopy only 34.2, MD −1.3 (−3.9 to 1.3); both surgical groups only a small benefit over no treatment (2.8 points vs decompression)**
- ☐ **FIMPACT (24 months)** (Paavola 2018 BMJ) [H]: **ASD vs diagnostic arthroscopy: pain VAS −4.6 (−11.3 to 2.1) at rest, −9.0 (−18.1 to 0.2) on activity — no clinically relevant difference**
- ☐ **Decompression vs exercise — 12 years** (Ketola 2017 Bone Joint J) [M]: **no significant difference in pain between surgery and exercise**
- ☐ **MID / PASS after subacromial surgery** (Kanto 2021 BMC Med Res Methodol) [M]: **MID: pain on activity 20 mm, Constant 10 points, SST 1.5; PASS: Constant 80, SST 11**

5 · FROZEN SHOULDER SURGICAL TRIALS (UK FROST & KIVIMÄKI)

RANGAN 2020 LANCET

- ☐ **Frozen shoulder — UK FROST (n = 503)** (Rangan 2020 Lancet) [H]: **12-month OSS: capsular release 40.3, MUA 38.3, early structured physio 37.2; all differences below target → none clinically superior; capsular release 8 serious adverse events, MUA 2; MUA most cost-effective**
- ☐ **MUA + home exercise vs home exercise** (Kivimäki 2007 J Shoulder Elbow Surg) [M]: **no difference in pain or working ability at any time; small ROM gains with MUA**
- ☐ **Arthroplasty & surgical precautions: POST-OP RESTRICTIONS ARE SURGEON-DIRECTED** [E]

6 · OPERATIVE PROCEDURE & SURGEON-DIRECTED PROTOCOL LOG

WRITE-IN

Procedure / Tear Size / Fixation _____ **Surgery Date** _____ **Operating Surgeon** _____ **Prior Injection < 1 m?** ☐ No ☐ Yes

Surgeon-Directed Restrictions & Milestones _____

OPERATIVE FINDINGS · SURGEON-DIRECTED RESTRICTIONS · POST-SURGICAL REHABILITATION NOTES

Clinician _____ **Date** _____

1 · ESWT & BARBOTAGE (CALCIFIC VS NON-CALCIFIC)

BANNURU · SURACE · MOOSMAYER

- ☐ **BESS on ESWT** Kulkarni 2015 (Guideline): **"Extracorporeal shock wave therapy may improve pain in calcific tendonitis."**
- ☐ **ESWT, calcific tendinitis (MA, 28 RCTs)** (Bannuru 2014 Ann Intern Med) [M]: **high-energy ESWT better than placebo for pain, function and calcium resorption in calcific tendinitis; no difference vs placebo in non-calcific tendinitis**
- ☐ **ESWT (Cochrane, 32 trials)** (Surace 2020 Cochrane) [M]: **vs placebo at 3 months: pain 0.78 better (0.17–1.4) on 0–10, below the 1.5 clinically important change; function 7.9 better (1.6–14), below 10 → benefit may not be clinically important**
- ☐ **Barbotage (lavage) for calcific tendinopathy (RCT, n = 218)** (Moosmayer 2023 BMJ) [H]: **OSS at 4 months: lavage + steroid vs sham 0.2 (−2.3 to 2.8); none superior to sham at 24 months**

2 · ELECTROTHERAPY (COCHRANE) & LAFRANCE 2022 CPG

PAGE 2016 · LAFRANCE 2022

- ☐ **Electrotherapy (Cochrane, 47 trials)** (Page 2016 Cochrane) [L]: **no meta-analysis possible; pulsed ultrasound for calcific tendinitis (1 trial, 61 participants, low quality): pain MD −8.60 (−13.48 to −3.72) on a 52-point scale**
- ☐ **JOSPT/Quebec CPG 2022 (cuff disorders)** Lafrance 2022 (Guideline): **"Clinicians should prescribe an active and task-oriented rehabilitation program (exercises and education)"**; **"Subacromial decompression is not recommended to treat rotator cuff tendinopathy"**; injections **"may be useful to reduce pain in the short term"**

3 · CORTICOSTEROID INJECTIONS & HYDRODILATATION (CUFF TENDINOSIS VS FROZEN SHOULDER)

MOHAMADI · CHALLOUMAS · CARETTE · KELLEY · BUCHBINDER

- ☐ **Corticosteroid injection, cuff tendinosis (MA)** (Mohamadi 2017 Clin Orthop Relat Res) [M]: **no better than placebo at 3 months; small transient relief at 4–8 weeks (SMD 0.52); at least 5 treated for 1 to benefit** (see p5 Forsythe 2019 if repair planned within 1 month: infection **1.3% vs 0.8%, OR 1.7, 1.0–2.9**)
- ☐ **Frozen shoulder treatments (NMA, 65 studies)** (Challoumas 2020 JAMA Netw Open) [M]: **only intra-articular corticosteroid clinically superior short term: pain vs placebo MD −1.0 (−1.5 to −0.5), vs physio −1.1 (−1.7 to −0.5) VAS**
- ☐ **Frozen shoulder: injection ± physio (RCT)** (Carette 2003 Arthritis Rheum) [M]: **injection + home exercise effective; adding supervised physio gives faster ROM gain; physio alone of limited efficacy; all groups similar at 12 months**
- ☐ **JOSPT CPG — injection** Kelley 2013 (Guideline): **"Intra-articular corticosteroid injections combined with shoulder mobility and stretching exercises are more effective in providing short-term (4-6 weeks) pain relief and improved function compared to shoulder mobility and stretching exercises alone." (strong evidence)**
- ☐ **Hydrodilatation / arthrographic distension (Cochrane)** (Buchbinder 2008 Cochrane) [L]: **1 low-risk-of-bias trial: distension with saline + steroid better than placebo at 3 weeks (NNTB 2 pain, 3 function)**

4 · ELECTROPHYSICAL AGENT & INJECTION PARAMETER LOG (BLANK WRITE-IN FIELDS)

PARAMETER LOG

RX	MODALITY / INJECTION TYPE	TARGET SITE & GUIDANCE	DOSE / PARAMETERS	DATE / WINDOW	RESPONSE & EXERCISE PAIRING
☐	Intra-articular steroid + mobility/stretching				
☐	Subacromial injection (short-term window)				
☐	Hydrodilatation (saline + steroid)				
☐	High-energy ESWT (calcific tendinitis)				
☐					

INJECTION TIMING · SHORT-TERM WINDOW PAIRING WITH EXERCISE · SHARED DECISION-MAKING NOTES

Clinician _____ Date _____

1 · ROTATOR CUFF MANUAL THERAPY

PAGE 2016 · DESJARDINS-CHARBONNEAU

- ☐ **MT + exercise for cuff disease (Cochrane, 60 trials)** (Page 2016 Cochrane) [M]: **vs placebo at 22 weeks: pain MD 6.8 (−0.70 to 14.30), function MD 7.1 (0.30 to 13.90) on 100-point scales — may not be clinically important**
- ☐ **MT for cuff tendinopathy (MA)** (Desjardins-Charbonneau 2015 JOSPT) [L]: **pain MD 1.1 (0.6–1.6) on 10-cm VAS vs placebo/added; MT added to exercise: 1.0 (0.7–1.4); effect on function unclear**

2 · FROZEN SHOULDER MOBILISATION VS SUPERVISED NEGLECT

PAGE 2014 · VERMEULEN · DIERCKS

- ☐ **MT + exercise for frozen shoulder (Cochrane)** (Page 2014 Cochrane) [M]: **at 7 weeks probably less improvement than glucocorticoid injection (pain MD 26 points, 15–37, on 100)**
- ☐ **High-grade vs low-grade mobilisation** (Vermeulen 2006 Phys Ther) [M]: **HGMT more effective for mobility and disability; overall differences small**
- ☐ **Supervised neglect vs intensive stretching** (Diercks 2004 J Shoulder Elbow Surg) [L]: **Constant ≥ 80 at 24 months: 89% supervised neglect vs 63% intensive; 64% within 12 months**

3 · THORACIC SMT & CERVICOTHORACIC CPR

KARDOUNI · PEEK · MINTKEN

- ☐ **Thoracic SMT vs sham (SIS)** (Kardouni 2015 JOSPT) [M]: **both groups improved (NPRS 1.2, Penn 9.1) with no difference between real and sham**
- ☐ **Thoracic manual therapy (SR)** (Peek 2015 J Man Manip Ther) [L]: **3 RCTs: reduced pain and disability vs usual care up to 52 weeks**
- ☐ **Cervicothoracic manipulation CPR** (Mintken 2010 Phys Ther) [L]: **≥ 3 of 5 variables: success 61% → 89% (LR+ 5.3); unvalidated single-arm study**

4 · DRY NEEDLING, KINESIO TAPING & STRETCHING

NAVARRO-SANTANA · CELIK · MCCLURE

- ☐ **Dry needling (MA)** (Navarro-Santana 2021 Phys Ther) [M]: **pain MD −0.49 (−0.84 to −0.13) — small, short term; disability MD −9.99 (−15.97 to −4.01) — low quality**
- ☐ **Kinesio taping (MA, 14 studies)** (Celik 2020 Clin Rehabil) [M]: **no better than sham for pain (MD −0.77, −1.77 to 0.22) or function → no firm evidence of benefit**
- ☐ **Posterior shoulder tightness** (McClure 2007 JOSPT) [L]: **cross-body stretch +20.0° IR (SD 12.9) vs control +5.9° (P = .009); sleeper stretch +12.4° (not significant)**

5 · MANUAL THERAPY, STRETCHING & PAIRED EXERCISE LOG

WRITE-IN LOG

RX	TECHNIQUE / STRETCH	TARGET JOINT / IRRITABILITY MATCH	IMMEDIATE PRE/POST RESPONSE	PAIRED ACTIVE EXERCISE (P8)
☐	**Glenohumeral mobilisation (low vs high grade)**			
☐	**Cross-body stretch (McClure 2007)**			
☐	**Thoracic / cervicothoracic mobilisation / SMT**			
☐	**Soft-tissue / myofascial / dry needling adjunct**			

6 · PRE-MOBILISATION, MANIPULATION & NEEDLING SAFETY SCREEN

CONTRAINDICATION GATE

- ☐ **BESS/BOA red flags excluded:** suspected infected joint, unreduced dislocation, tumour/malignancy & acute traumatic tear screened (p1) (Guideline)
- ☐ **Posterior dislocation & fracture excluded:** trauma with locked IR or greater-tuberosity fracture cleared (p1) [L]
- ☐ **Stage & irritability matched:** avoid aggressive overstretching in high-irritability frozen shoulder (Diercks 2004 [L]; Kelley 2013, Guideline)
- ☐ **Needling & thoracic safety:** pneumothorax / lung field landmarks, anticoagulants, and neurovascular status checked [E]

CONSENT OBTAINED FOR: ☐ GH / scapular mobilisation ☐ Thoracic manipulation / SMT ☐ Dry needling / soft-tissue | ☐ Risks & alternatives discussed ☐ Patient may decline **Signature** _____

MANUAL THERAPY CLINICAL REASONING · IRRITABILITY MATCHING · IMMEDIATE RE-TEST NOTES

Clinician _____ **Date** _____

Exercise Therapy, Return to Sport & 12-Week Loading Log

Patient _____

1 · ROTATOR CUFF / RCRSP EXERCISE TRIALS & DOSING PRINCIPLES

GRASP · HOLMGREN · SELF · INGWERSEN · SMITH · SAITO

- ☐ **GRASP (n = 708)** (Hopewell 2021 Lancet) [H]: **progressive exercise vs one best-practice advice session: SPADI over 12 months −0.66 (99% CI −4.52 to 3.20) — no difference; injection vs no injection −1.11 (−4.47 to 2.26)**
- ☐ **Specific exercise (Holmgren)** (Holmgren 2012 BMJ) [H]: **Constant-Murley +24 vs +9 points (MD 15, 8.5–20.6); surgery chosen by 20% vs 63% (OR 7.7, 3.1–19.4)**
- ☐ **Single self-managed exercise (SELF)** (Littlewood 2016 Clin Rehabil) [M]: **SPADI at 3 months 32.4 vs 30.7 usual physio; adjusted MD 3.2 (−6.0 to 12.4) — comparable**
- ☐ **Heavy vs low load (RCT, n = 100)** (Ingwersen 2017 Orthop J Sports Med) [M]: **DASH adjusted difference −1.37 (−6.72 to 3.99) — no superiority of high-load**
- ☐ **Painful vs pain-free exercise (MA)** (Smith 2017 BJSM) [M]: **short-term pain SMD −0.27 (−0.54 to −0.05) in favour of painful exercise; no difference later**
- ☐ **Scapular-focused exercise (MA, 6 RCTs)** (Saito 2018 Physiother Theory Pract) [M]: **short-term pain MD −0.88 (−1.19 to −0.58), function −11.31 (−17.20 to −5.41); no difference at follow-up**

2 · FULL-THICKNESS TEARS: EXERCISE VS REPAIR

KUKKONEN · MOOSMAYER

- ☐ **Non-traumatic cuff tears > 55 y (RCT, 180 shoulders)** (Kukkonen 2014 Bone Joint J) [H]: **Constant at 1 year: physio 74.1, acromioplasty 77.2, repair 77.9 (p = 0.34)**
- ☐ **Small/medium tears — 10 years** (Moosmayer 2019 JBJS Am) [H]: **repair better than physio by 9.6 Constant points (p = 0.002), ASES 15.7; crossovers to late surgery 10.0 points worse than primary repair**

3 · MDI WATSON PROGRAMME & FROZEN SHOULDER CPG

WARBY · KELLEY

- ☐ **MDI: Watson vs Rockwood programme** (Warby 2018 Am J Sports Med) [M]: **WOSI effect 11.1 (1.9–20.2) at 12 weeks; 12.6 (3.4–21.9) at 24 weeks — favours Watson**
- ☐ **Frozen shoulder education** Kelley 2013 (Guideline): **JOSPT CPG: educate on natural course, activity modification, and match stretching intensity to irritability**

4 · OVERHEAD THROWERS — GIRD, TOTAL ROTATIONAL MOTION (TRM) & EXTERNAL ROTATION

WILK 2011 · WILK 2015

- ☐ **Throwers — GIRD / TRM** (Wilk 2011 Am J Sports Med) [L]: **GIRD ≈ twice the injury rate but not significant (P = .17); total rotational motion deficit > 5° → higher injury rate**
- ☐ **Throwers — ER** (Wilk 2015 Am J Sports Med) [L]: **insufficient ER (< 5° more ER on throwing side): 2.2× disabled-list risk (1.2–4.1), 4.0× surgery (1.5–12.6); GIRD not significant**

5 · 12-WEEK SHOULDER EXERCISE & RETURN-TO-SPORT PROGRESSION LOG

12-WEEK LOG

Primary Programme Ticked: ☐ GRASP Best-Practice / Progressive ☐ Holmgren Specific Cuff ☐ SELF Single Exercise ☐ Scapular-Focused (Saito) ☐ Watson MDI Programme ☐ Irritability-Matched Stretching (Kelley)

EXERCISE / REHABILITATION DOMAIN	PHASE I (BASELINE)	PHASE II	PHASE III	PHASE IV	WEEK 12 / RTS
1 · Primary Cuff / Self-Managed Exercise					
2 · Scapular Upward Rotation / Control					
3 · Posterior Shoulder / Mobility					
4 · Kinetic Chain / CKCUEST / Overhead					
5 · Symptom Response (NPRS)					

EXERCISE PRESCRIPTION DETAILS · LOAD PROGRESSION · RETURN-TO-SPORT CRITERIA NOTES

Clinician _____ Date _____

Longitudinal Session-by-Session Tracking Flowsheet

Patient _____

#	DATE	NPRS	ER 0°/90°	ELEV / IR	SAT/SSMP	EXERCISE & LOADING DELIVERED	MANUAL & ADJUNCTS	IRRITABILITY	ADHERENCE	INIT
1						GRASP | Holmgren | SELF | Scap | Watson | Stretch	GH Mob | Cross-Body | Thoracic | DN	Low | Mod | High		
2						GRASP | Holmgren | SELF | Scap | Watson | Stretch	GH Mob | Cross-Body | Thoracic | DN	Low | Mod | High		
3						GRASP | Holmgren | SELF | Scap | Watson | Stretch	GH Mob | Cross-Body | Thoracic | DN	Low | Mod | High		
4						GRASP | Holmgren | SELF | Scap | Watson | Stretch	GH Mob | Cross-Body | Thoracic | DN	Low | Mod | High		
5						GRASP | Holmgren | SELF | Scap | Watson | Stretch	GH Mob | Cross-Body | Thoracic | DN	Low | Mod | High		
6						GRASP | Holmgren | SELF | Scap | Watson | Stretch	GH Mob | Cross-Body | Thoracic | DN	Low | Mod | High		
7						GRASP | Holmgren | SELF | Scap | Watson | Stretch	GH Mob | Cross-Body | Thoracic | DN	Low | Mod | High		
8						GRASP | Holmgren | SELF | Scap | Watson | Stretch	GH Mob | Cross-Body | Thoracic | DN	Low | Mod | High		
9						GRASP | Holmgren | SELF | Scap | Watson | Stretch	GH Mob | Cross-Body | Thoracic | DN	Low | Mod | High		
10						GRASP | Holmgren | SELF | Scap | Watson | Stretch	GH Mob | Cross-Body | Thoracic | DN	Low | Mod | High		
11						GRASP | Holmgren | SELF | Scap | Watson | Stretch	GH Mob | Cross-Body | Thoracic | DN	Low | Mod | High		
12						GRASP | Holmgren | SELF | Scap | Watson | Stretch	GH Mob | Cross-Body | Thoracic | DN	Low | Mod | High		

DISCHARGE & OUTCOME MILESTONE SUMMARY (CHANGE VS VERIFIED MCID)

FINAL OUTCOME

SHOULDER PROM & VERIFIED THRESHOLD	INITIAL	DISCHARGE	Δ	MCID MET
SPADI (MDC90 18.1; MCID 8.0 and 13.1 points)				☐
QuickDASH / DASH (MCID 8.0; MA 11.97 / 11.00; FS 10.2)				☐
ASES (MDC 9.7; MCID 6.4 points)				☐
Oxford Shoulder Score (MCIC 6.50 / 8.42; "at least eight points")				☐
Constant (MCID 10.4; MID 10, PASS 80)				☐
NPRS (MCID 1.1) & PSFS (write-in [E])				☐

Discharge: ☐ Goals & MCID met ☐ Self-management / RTS ☐ Orthopaedic / surgical review ☐ Non-responder

OBJECTIVE RE-TEST MILESTONES

SERIAL RE-TEST

OBJECTIVE MEASURE	BASE	MID	D/C	MET
Passive & Active ER (0° / 90°)				☐
Total Rotational Motion (ER + IR)				☐
Isometric Strength / HHD Symmetry				☐
CKCUEST (≥ 3 touches = real change)				☐
Scapular Dyskinesis / SAT / SRT				☐
Special Test Cluster Status				☐

HOME EXERCISE MAINTENANCE · OVERHEAD / RTS PLAN · DISCHARGE SUMMARY

Clinician / Surgeon follow-up _____ Signature _____ Date _____

Discharging Clinician _____ Sig. _____